

Subhadeep Dhar

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SUMMARY

Full-Stack Software Engineer and MCA candidate with hands-on experience building and deploying production web and mobile ecosystems. Proficient in TypeScript, Next.js, Node.js, React Native, and Flutter. Experienced in designing scalable REST APIs, role-based dashboards, on-device machine learning workflows, and automated scraping pipelines. Proven track record of translating complex organizational requirements into optimized, low-latency software solutions.

EXPERIENCE

Research Intern — Centre for Health Policy (CHP)

Dec 2024 – Feb 2025

Asian Development Research Institute (ADRI)

Patna, Bihar, India

- Supported a Government of Bihar public health initiative aiming to strengthen the IHIP-IDSP disease surveillance framework by analyzing large data pipelines to identify regional reporting gaps and data inconsistencies.
- Performed data collection, preprocessing, and analytical reporting to identify inconsistencies and operational gaps in disease surveillance datasets.
- Engineered an internal workforce management dashboard using a PHP stack to automate staff availability tracking and weekly scheduling, reducing administrative overhead for ADRI research personnel.

Full Stack Developer

Jul 2024 – Feb 2025

SHIELD Tech Infosec Solutions

Gangtok, Sikkim, India

- Why & Ecosystem Impact:** Developed full-stack web solutions for local hospitality businesses, homestays, and regional travel agencies to lower onboarding barriers and expand the digital footprint of Sikkim's local tourism economy.
- How & Technical Execution:** Implemented performant, fluid user interfaces backed by decoupled Express.js REST APIs and MySQL relational databases; designed modular content management pipelines to securely handle structured customer queries.

PROJECTS

Dooars Tutors — Regional Skill Discovery Platform | Live | GitHub

Mar 2025 – Present

Founder / Lead Full-Stack Engineer

TypeScript, Next.js 16, Node.js, MongoDB, Redis

- Architected a regional discovery platform to resolve severe educational access inequalities in the Alipurduar region, eliminating digital visibility monopolies held by prominent institutions and providing hyper-local tutors an equal platform to be discovered.
- Engineered a full-stack TypeScript monorepo using npm workspaces to unify a Next.js 16/React 19 client with a modular Express backend; built a high-performance data infrastructure featuring MongoDB geospatial indexing, Google Maps API, and an automated data-scraping pipeline via Playwright and Cheerio.
- Centralized regional educational options by onboarding 40+ active tutors to seamlessly connect with over 500+ active students; maintained 100% platform availability and eliminated user drop-off by combining a Redis caching layer with automated UptimeRobot cron polling to completely bypass free-tier container cold-starts.

Poultry Disease Detection App — Mobile Veterinary AI Prototype | GitHub

2026

Lead Mobile & ML Developer

Flutter, Dart, TensorFlow Lite, MobileNet V2

- Conceptualized and developed a veterinary AI application to mitigate severe livestock and production losses for poultry farmers in Bihar, focusing on the early-stage identification of devastating regional outbreaks like Newcastle disease and Aflatoxicosis.
- Engineered a cross-platform Flutter application utilizing a modular, service-based architecture; integrated an on-device TensorFlow Lite pipeline with a MobileNet V2 neural network to extract image feature vectors and execute local Cosine Similarity matching against pre-computed JSON datasets.
- Delivered an entirely offline, zero-latency diagnostic utility designed for remote, disconnected rural environments; implemented a strict 0.75 model confidence threshold for prediction validation alongside automated, shareable PDF health report generation.

TECHNICAL SKILLS

Programming Languages: TypeScript, JavaScript, Dart, Python, C++, SQL, Java

Mobile & Frontend Development: React, React Native, Flutter, Expo, Expo Router, Next.js, Tailwind CSS, Framer Motion, shadcn/ui

Backend & APIs: Node.js, Express.js, REST APIs, FastAPI, TensorFlow Lite, Firebase, Firestore, Google Maps API

Databases & Cloud: MongoDB, MongoDB Atlas, Mongoose, MySQL, Redis, Firebase Storage

DevOps & Tools: Git, Docker, GitHub Actions (CI/CD), Postman, EAS Build, Expo CLI, Vercel, Render, Upstash

EDUCATION

Manipal Institute of Technology (MIT)

Karnataka, India

Master of Computer Applications (MCA)

Aug 2025 – May 2027

Sikkim Manipal Institute of Technology

Sikkim, India

Bachelor of Computer Applications (BCA) — Specialization in Data Science

Aug 2022 – May 2025